

UET University Peshawar

Abbottabad Campus



Lab Report No: 05

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Date: 18 June 2026

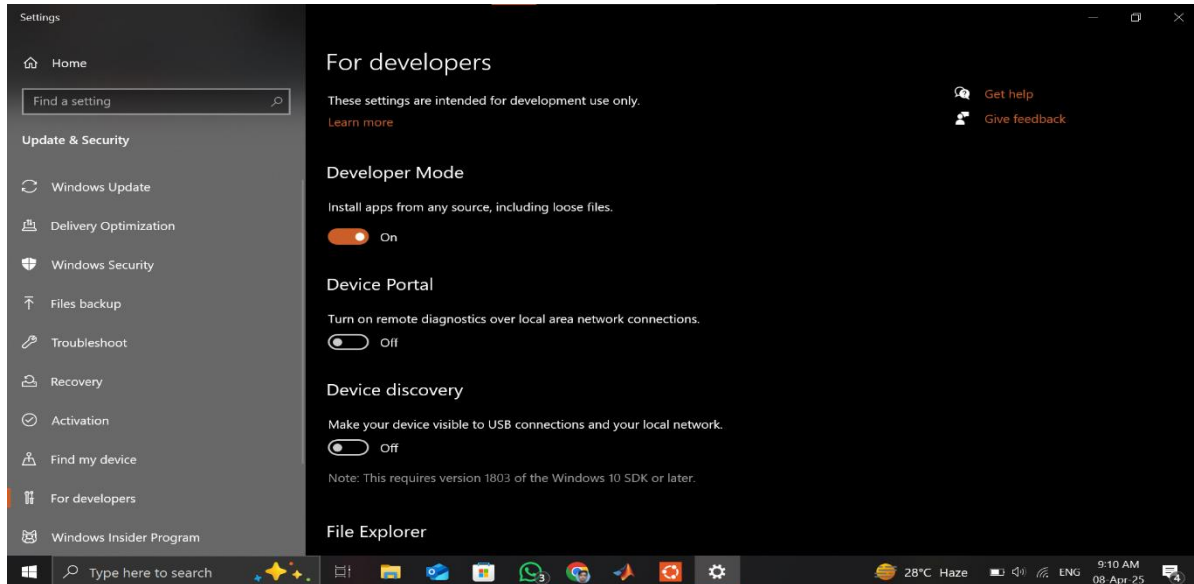
Department of Software Engineering
University of Engineering & Technology, Abbottabad Campus

Step by Step Installation of NS2 on Windows 10/11

STEP 1

Turn on Developer Mode

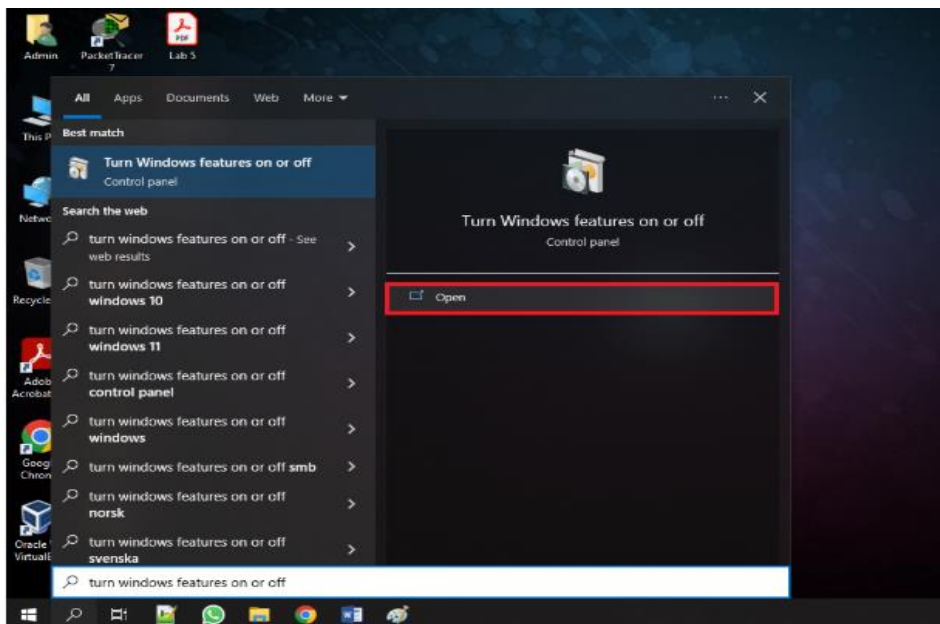
Open search (Windows + S) and type “developer settings” and click on Open as shown below:



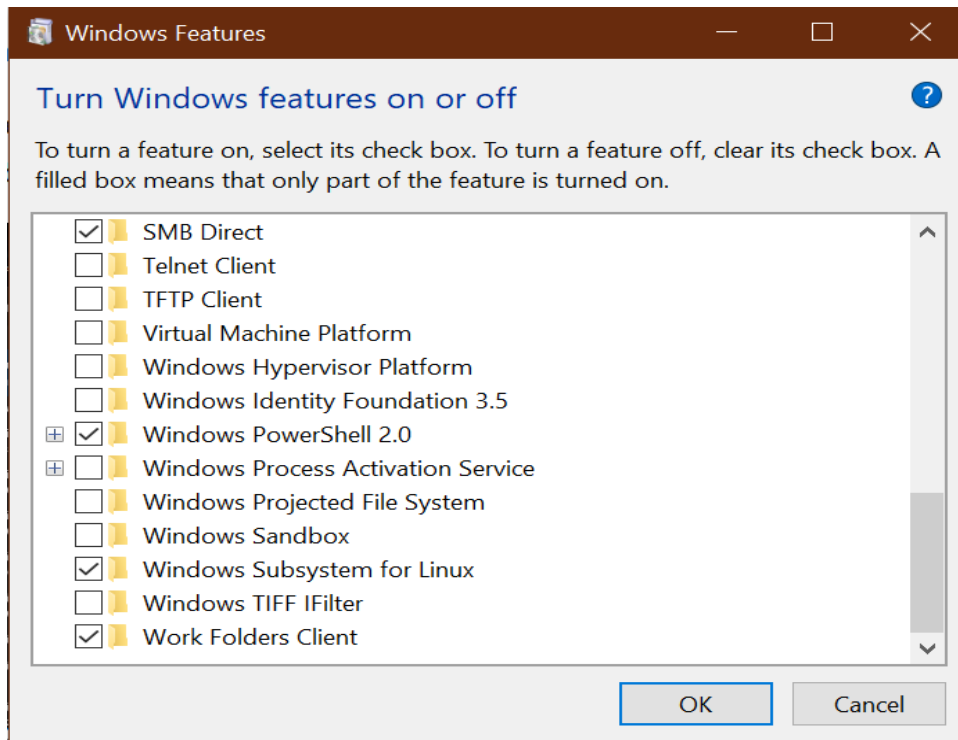
STEP 2

Turn on WSL (Windows Subsystem for Linux)

Open search and type “turn windows features on or off” and click on Open as shown below:



Scroll down the options and check the item named “Windows Subsystem for Linux” and click on OK as shown below:

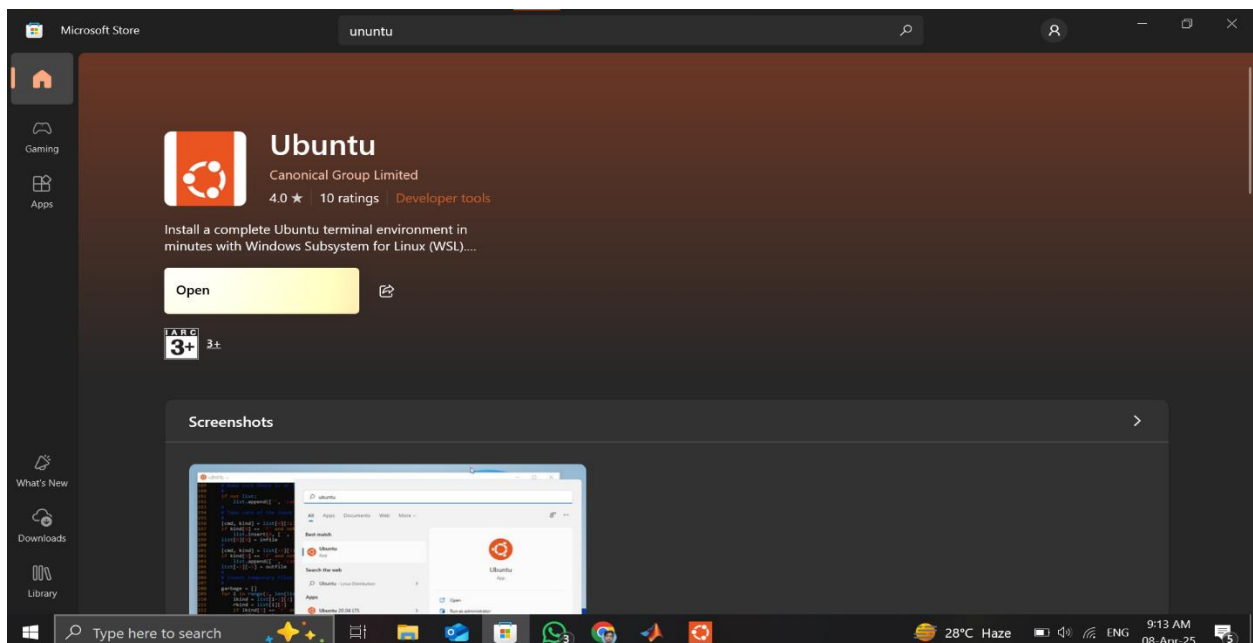


STEP 3

Install Ubuntu from Microsoft Store.

Click on search and type “Microsoft Store” and click on Open.

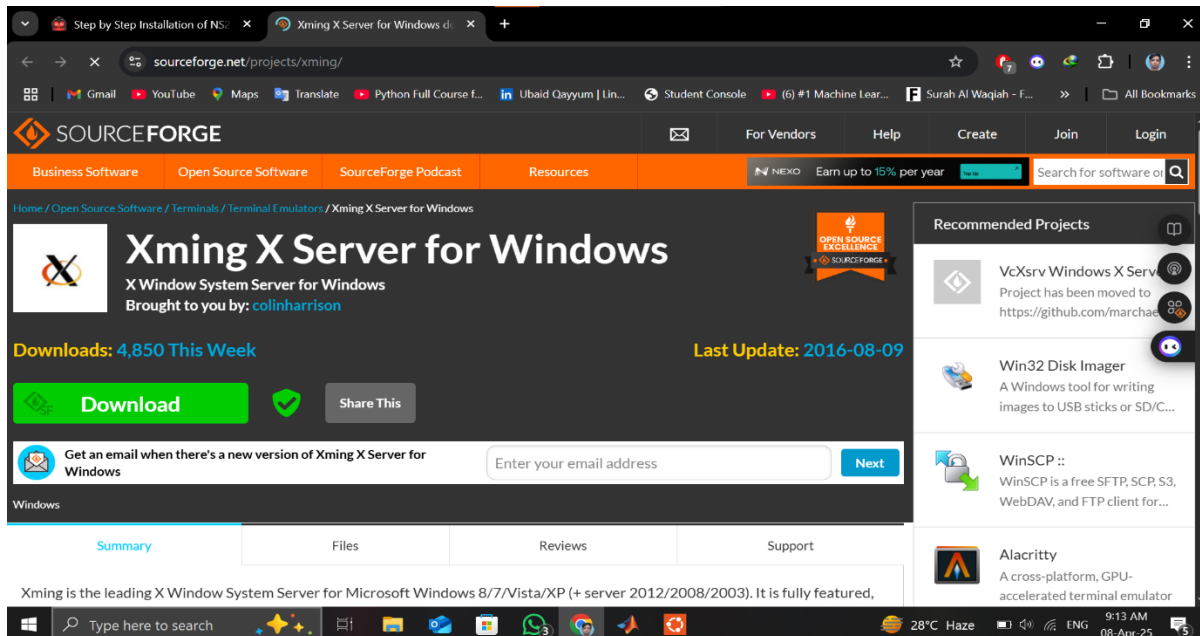
In the search bar at the top, type “Ubuntu” and click on “Install” button as shown below:



STEP 4

Install Xming.

Open Google in a browser and type “Xming Download” or go to this URL <https://sourceforge.net/projects/xming/> and click on the “Download” button.



After downloading Xming, install it.

STEP 5

Open search and type “Ubuntu” and click on Open.

As it is the first time you are opening Ubuntu, it will ask for Username and Password. Give a username and password and hit enter key. Now you should see a prompt as shown below:

Type the following commands one by one and hit enter after each command:

```
ubaid@ubaid:~$ sudo apt update
[sudo] password for ubaid:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [741 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [141 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [8956 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [829 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [180 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [259 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [173 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [989 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [218 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [151 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [13.5 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1051 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [265 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [365 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [26.0 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [892 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [181 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
```

```

ubaid@Ubaid:~$ sudo apt-get install ns2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ibverbs-providers libibverbs1 libnl-3-200 libnl-route-3-200 libotcl1
  libpcap0.8t64 libtcl8.6 libtclcl1 libtk8.6 libxft2 libxss1
Suggested packages:
  tcl8.6 tk8.6 gnuplot
The following NEW packages will be installed:
  ibverbs-providers libibverbs1 libnl-3-200 libnl-route-3-200 libotcl1
  libpcap0.8t64 libtcl8.6 libtclcl1 libtk8.6 libxft2 libxss1 ns2
0 upgraded, 12 newly installed, 0 to remove and 144 not upgraded.
Need to get 5316 kB of archives.
After this operation, 24.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-route-3-200 amd64 3.7.0-0.3build1.1 [189 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libibverbs1 amd64 50.0-2build2 [67.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 ibverbs-providers amd64 50.0-2build2 [374 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libpcap0.8t64 amd64 1.10.4-4.1ubuntu3 [151 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libtcl8.6 amd64 8.6.14+dfsg-1build1 [988 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 libotcl1 amd64 1.14+dfsg-8 [24.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libtclcl1 amd64 1.20-10 [65.2 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libxft2 amd64 2.3.6-1build1 [45.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 libxss1 amd64 1:1.2.3-1build3 [7204 B]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 libtk8.6 amd64 8.6.14-1build1 [779 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 ns2 amd64 2.35+dfsg-5build2 [2569 kB]
Fetched 5316 kB in 5s (1111 kB/s)
Selecting previously unselected package libnl-3-200:amd64.
(Reading database ... 40794 files and directories currently installed.)
Preparing to unpack .../00-libnl-3-200_3.7.0-0.3build1.1_amd64.deb ...
Unpacking libnl-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package libnl-route-3-200:amd64.
Preparing to unpack .../01-libnl-route-3-200_3.7.0-0.3build1.1_amd64.deb ...Unpacking libnl-route-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package libibverbs1:amd64.
Preparing to unpack .../02-libibverbs1_50.0-2build2_amd64.deb ...
Unpacking libibverbs1:amd64 (50.0-2build2) ...
Selecting previously unselected package ibverbs-providers:amd64.
Preparing to unpack .../03-ibverbs-providers_50.0-2build2_amd64.deb ...
Unpacking ibverbs-providers:amd64 (50.0-2build2) ...

ubaid@Ubaid:~$ sudo apt-get install gedit
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf aspell aspell-en bubblewrap
  dictionaries-common docbook+xml emacs-en-common enchant-2 gedit-common
  gir1.2-amtk-5 gir1.2-atk-1.0 gir1.2-freedesktop gir1.2-gdkpixbuf-2.0
  gir1.2-gtk-3.0 gir1.2-gtksource-300 gir1.2-harfbuzz-0.0 gir1.2-pango-1.0
  gir1.2-peas-1.0 gir1.2-tepl-6 glib-networking glib-networking-common
  glib-networking-services gstreamer1.0-gl gstreamer1.0-plugins-base
  gstreamer1.0-plugins-good gstreamer1.0-x hunspell-en-us libaa1
  libadwaita-1-0 libasound2-data libasound2t64 libaspell15 libasyncns0
  libatomic1 libavc1394-0 libcacat0 libcairo-script-interpreter2
  libcdparanoia0 libdv4t64 libenchant-2-2 libevdev2 libflac12t64
  libgedit-amtk-5-0 libgedit-amtk-5-common libgedit-gtksourceview-300-0
  libgedit-gtksourceview-300-common libgles2 libgraphene-1.0-0
  libgspell-1-2 libgspell-1-common libgstreamer-glib1.0-0
  libgstreamer-plugins-base1.0-0 libgstreamer-plugins-good1.0-0 libgtk-4-1
  libgtk-4-bin libgtk-4-common libgtk-4-media-gstreamer libgudev-1.0-0
  libhandy-1-0 libharfbuzz-gobject0 libharfbuzz-icu0 libhunspell-1.7-0
  libhyphen0 libiec61883-0 libjavascriptcoregtk-4.1-0 libjson-glib-1.0-0
  libjson-glib-1.0-common libmanette-0.2-0 libmp3lame0 libmpeg123-0t64
  libncurses6 libogg0 libopus0 liborc-0.4-0t64 libpangoxft-1.0-0
  libpeas-1.0-0 libpeas-common libpipewire-0.3-0t64 libpipewire-0.3-common
  libproxy1v5 libpulse0 libraw1394-11 libsecret-1-0 libsecret-common
  libshout3 libsndfile1 libsoup-3.0-0 libsoup-3.0-common
  libspa-0.2-modules libspeek1 libtag1v5 libtag1v5-vanilla libtepl-6-4
  libtepl-common libtheora0 libtwolame0 libv4l-0t64 libv4lconvert0t64
  libvisual-0.4-0 libvorbis0a libvorbisenc2 libvpx9 libwavpack1
  libwebkit2gtk-4.1-0 libwebpdemux2 libwebpmux3
  libwebRTC-audio-processing1 libwoff1 libxslt1.1 libxv1 libyelp0
  python3-cairo python3-gi-cairo sgml-data xdg-dbus-proxy
  xdg-desktop-portal xdg-desktop-portal-gtk yelp yelp-xsl zenity
  zenity-common
Suggested packages:
  aspell-doc spellutils wordlist docbook docbook-dsssl docbook-xsl
  docbook-defguide gedit-plugins gvfs hunspell openoffice.org-hunspell
  | openoffice.org-core alsa-utils libasound2-plugins libdv-bin oss-compat
  libenchant-2-voikko libvisual-0.4-plugins opus-tools pipewire pulseaudio
  libraw1394-doc speex gstreamer1.0-libav gstreamer1.0-plugins-bad

```

```

ubaïd@Ubaïd:~$ sudo apt install tcl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  tcl8.6
Suggested packages:
  tcl-tclreadline
The following NEW packages will be installed:
  tcl tcl8.6
0 upgraded, 2 newly installed, 0 to remove and 144 not upgraded.
Need to get 18.9 kB of archives.
After this operation, 63.5 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 tcl8.6 amd64 8.6.14+dfsg-1build1 [14.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 tcl amd64 8.6.14build1 [4124 B]
Fetched 18.9 kB in 1s (13.3 kB/s)
Selecting previously unselected package tcl8.6.
(Reading database ... 46824 files and directories currently installed.)
Preparing to unpack .../tcl8.6_8.6.14+dfsg-1build1_amd64.deb ...
Unpacking tcl8.6 (8.6.14+dfsg-1build1) ...
Selecting previously unselected package tcl.
Preparing to unpack .../tcl_8.6.14build1_amd64.deb ...
Unpacking tcl (8.6.14build1) ...
Setting up tcl8.6 (8.6.14+dfsg-1build1) ...
Setting up tcl (8.6.14build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubaïd@Ubaïd:~$

```

STEP 6

Changing the location to working directory.

For this example, for working directory name is “ns2” and it is located on my “Desktop”. The complete path to “ns2” directory on my PC is “C:\Users\Admin\Desktop\ns2”.

To change the location to the above directory, type the following command and hit Enter key.

```
cd /mnt/c/Users/Admin/Desktop/ns2
```

STEP 7

Opening gedit and typing the program.

Type the following command in the terminal (black window) and hit Enter key.

```

ubaïd@Ubaïd:~$ cd /mnt/c/Users/PMLS/Desktop
ubaïd@Ubaïd:/mnt/c/Users/PMLS/Desktop$ cd ns2
ubaïd@Ubaïd:/mnt/c/Users/PMLS/Desktop/ns2$ gedit ex1.tcl

(gedit:2204): Gtk-WARNING **: 11:29:15.967: cannot open display:
ubaïd@Ubaïd:/mnt/c/Users/PMLS/Desktop/ns2$ export DISPLAY=0:0
ubaïd@Ubaïd:/mnt/c/Users/PMLS/Desktop/ns2$ _

```

You should be able to see a blank gedit window as shown in below image.

Type the following sample program and save (CTRL + s) the file.

Code:

```
ex1.tcl
/mnt/c/Users/PMLS/Desktop/ns2

1 # Create global variables
2 set ns [new Simulator]
3
4 # Setting NAM trace
5 set namf [open wired1.nam w]
6 $ns namtrace-all $namf
7
8 # Open the trace file
9 set tracef [open wired1.tr w]
10 $ns trace-all $tracef
11
12 # Protocol setting (not used further in the script, just defined)
13 set proto rlm
14
15 # Setting the color values
16 $ns color 1 blue
17 $ns color 2 yellow
18 $ns color 3 red
19
20 # Creating client - router - end server node
21 set n0 [$ns node]
22 set n1 [$ns node]
23 set n2 [$ns node]
24
25 # Establish the link between the nodes
26 $ns duplex-link $n0 $n1 2Mb 100ms DropTail
27 $ns duplex-link $n1 $n2 200Kb 100ms DropTail
28
29 # Label the nodes
30 $ns at 0.0 "$n0 label Client1"
31 $ns at 0.0 "$n1 label Server"
32 $ns at 0.0 "$n2 label Client2"
33
34 # Set node colors (note: NS2 supports node coloring for NAM visualization, not custom shapes)
35 $ns at 0.1 "$n0 color blue"
```

Now close gedit window.

STEP 8

Running the program using “ns” command.

Type the following command in the terminal to run the program and see the output.

```
ubaid@Ubaid:/mnt/c/Users/PMLS/Desktop/ns2$ ns ex1.tcl
Opening NAM...
ubaid@Ubaid:/mnt/c/Users/PMLS/Desktop/ns2$ nam:
```

